

Thadomal Shahani Engineering College, Bandra
Department of Artificial Intelligence & Data Science

Name: Jyotirmoy Dutta				Subject: SEPM Lab
TE- 1	Batch: T13	Sem: 6	Roll No: 2301051	Date: 02-02-2026

Experiment No: 4

Aim: To understand Continuous Integration, install and configure Jenkins with Maven/Ant/Gradle to setup a build Job

Steps:

To install Jenkins following software packages are required

- 1) GIT (git-scm.com)
- 2) Notepad++ (<https://notepad-plus-plus.org/downloads/>)
- 3) Latest Java development kit (JDK)
- 4) Jenkins
- 5) Apache Maven (Optional)

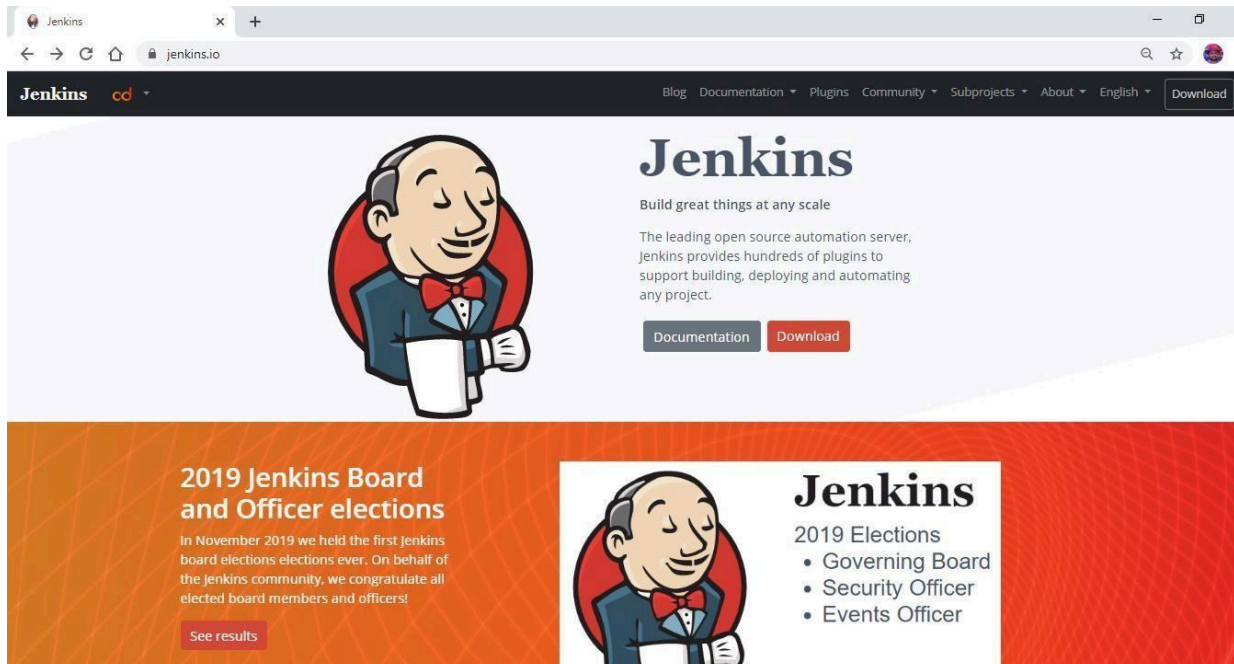
Step 1 -: Install GIT

Step 2 -: Install Notepad++

Step 3 -: Install Java

Step 4 -: Install Jenkins

Step 5 -: Install Maven



Jenkins

Build great things at any scale

The leading open source automation server, Jenkins provides hundreds of plugins to support building, deploying and automating any project.

Documentation Download

2019 Jenkins Board and Officer elections

In November 2019 we held the first Jenkins board elections ever. On behalf of the Jenkins community, we congratulate all elected board members and officers!

See results

Jenkins

2019 Elections

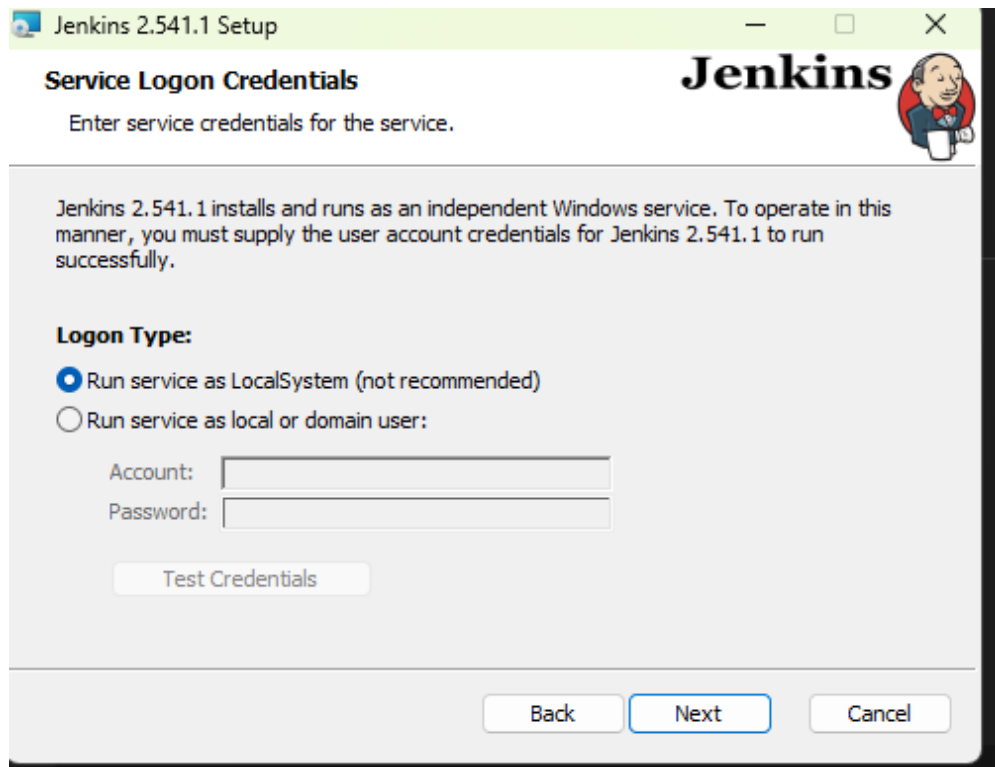
- Governing Board
- Security Officer
- Events Officer

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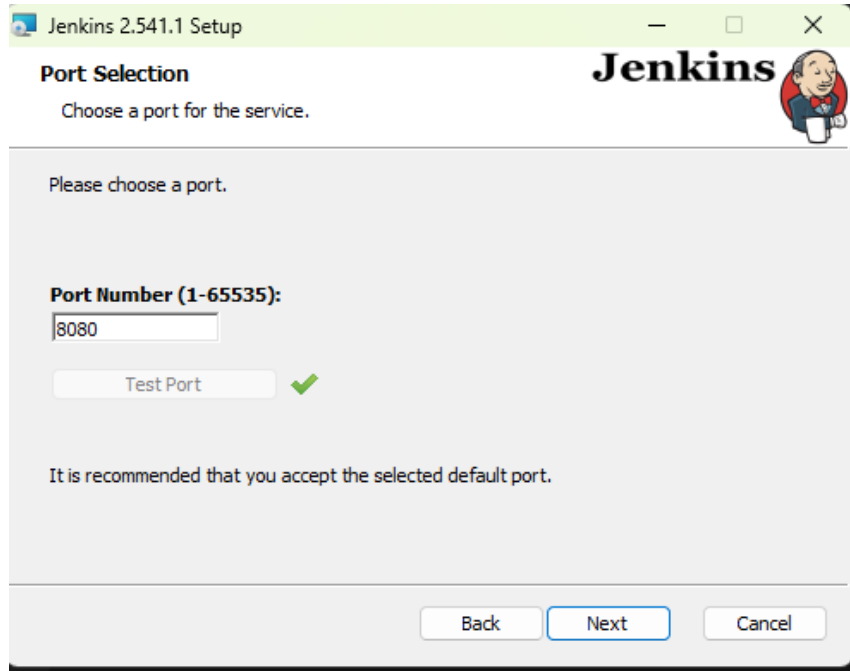
Jenkins is an open source automation tool written in Java with plugins built for Continuous Integration purpose. Jenkins is used to build and test your software projects continuously making it easier for developers to integrate changes to the project, and making it easier for users to obtain a fresh build. It also allows you to continuously deliver your software by integrating with a large number of testing and deployment technologies.

Output:

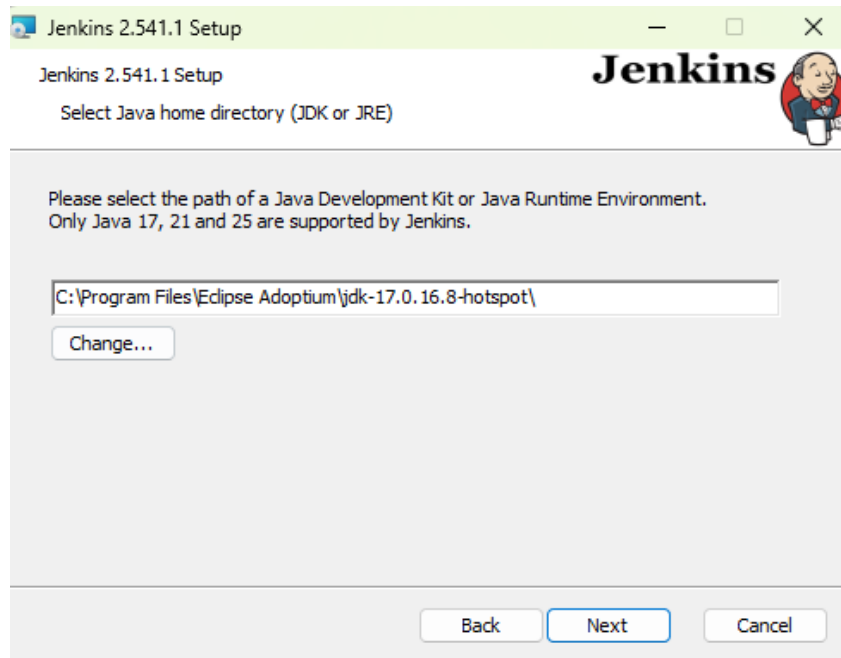


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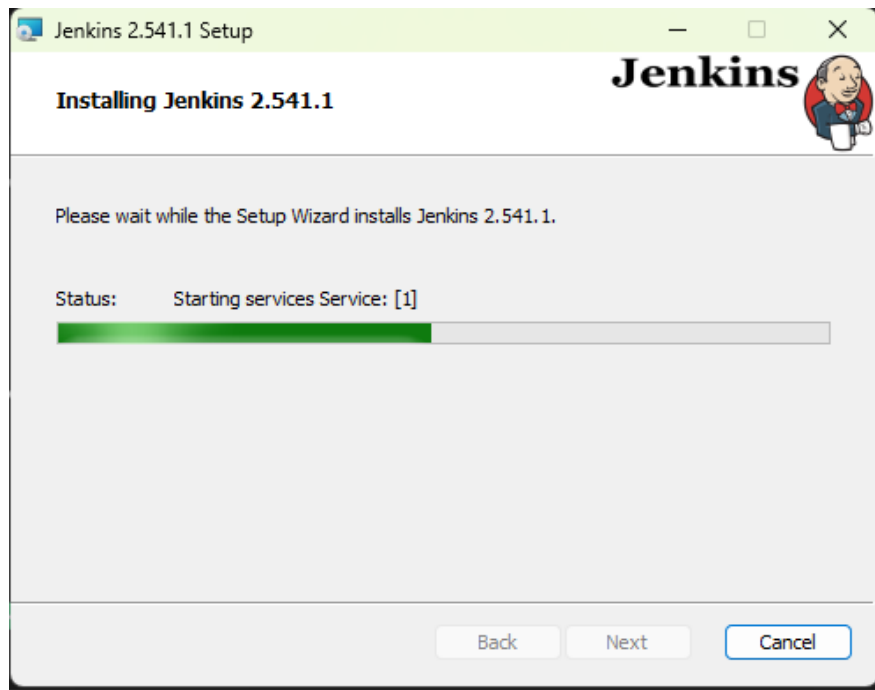
The screenshot shows the 'Port Selection' window of the Jenkins 2.541.1 Setup. The window has a title bar 'Jenkins 2.541.1 Setup' and a logo on the right. The main text says 'Choose a port for the service.' Below this, it says 'Please choose a port.' There is a text input field labeled 'Port Number (1-65535):' containing the value '8080'. To the right of the input field is a 'Test Port' button with a green checkmark next to it. At the bottom, there is a message: 'It is recommended that you accept the selected default port.' and three buttons: 'Back', 'Next', and 'Cancel'.



The screenshot shows the 'Select Java home directory (JDK or JRE)' window of the Jenkins 2.541.1 Setup. The window has a title bar 'Jenkins 2.541.1 Setup' and a logo on the right. The main text says 'Select Java home directory (JDK or JRE)'. Below this, it says 'Please select the path of a Java Development Kit or Java Runtime Environment. Only Java 17, 21 and 25 are supported by Jenkins.' There is a text input field containing the path 'C:\Program Files\Eclipse Adoptium\jdk-17.0.16-hotspot\' and a 'Change...' button below it. At the bottom, there are three buttons: 'Back', 'Next', and 'Cancel'.

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The screenshot shows the Jenkins Dashboard in a web browser. The left sidebar contains navigation links: 'New Item', 'Build History', 'Build Queue' (showing 'No builds in the queue'), and 'Build Executor Status' (showing '0/2'). The main area displays a table of builds with columns: 'S' (Status), 'W' (Icon), 'Name', 'Last Success', 'Last Failure', and 'Last Duration'. The table lists various builds including 'example0', 'Example1', 'example12', 'Example2', 'example_1', 'exp 1', 'exp 6', 'Hello_java', 'kannya', 'nisarg', 'pipeline demo', 'prototyoe1', and 'prototype'. Each row includes a status icon (green checkmark for success, red X for failure, or blue circle with dots for in progress) and a duration. A search bar and 'Add description' button are at the top right.

S	W	Name	Last Success	Last Failure	Last Duration
✓	☀	example0	10 mo #2	N/A	0.11 sec
✓	☀	Example1	10 mo #1	N/A	0.1 sec
✓	☀	example12	10 mo #5	N/A	0.1 sec
...	☀	Example2	N/A	N/A	N/A
✓	☁	example_1	2 yr 0 mo #5	2 yr 0 mo #4	0.11 sec
...	☀	exp 1	N/A	N/A	N/A
✓	☀	exp 6	5 days 21 hr #1	N/A	0.16 sec
✓	☀	Hello_java	11 mo #1	N/A	72 ms
✗	☁	kannya	11 mo #3	11 mo #10	0.11 sec
✓	☀	nisarg	11 mo #2	N/A	0.12 sec
✓	☁	pipeline demo	2 yr 11 mo #2	2 yr 11 mo #1	2.4 sec
...	☀	prototyoe1	N/A	N/A	N/A
✓	☀	prototype	10 mo #1	N/A	0.18 sec

The screenshot shows the Jenkins Console for a specific build. The breadcrumb navigation is 'Jenkins / T13_63 / #1'. The left sidebar has links: 'Status', 'Changes', 'Console Output' (selected), 'Edit Build Information', 'Delete build #1', and 'Next Build'. The main area is titled 'Console' and shows the build's execution log. The log indicates the build started by an unknown user, ran as SYSTEM, and executed a series of commands in a Windows environment, including a directory listing and an echo command. The build finished successfully.

```
Started by user unknown or anonymous
Running as SYSTEM
Building in workspace C:\ProgramData\Jenkins\jenkins\workspace\T13_63
[T13_63] $ cmd /c call C:\WINDOWS\TEMP\jenkins3599814823258889476.bat

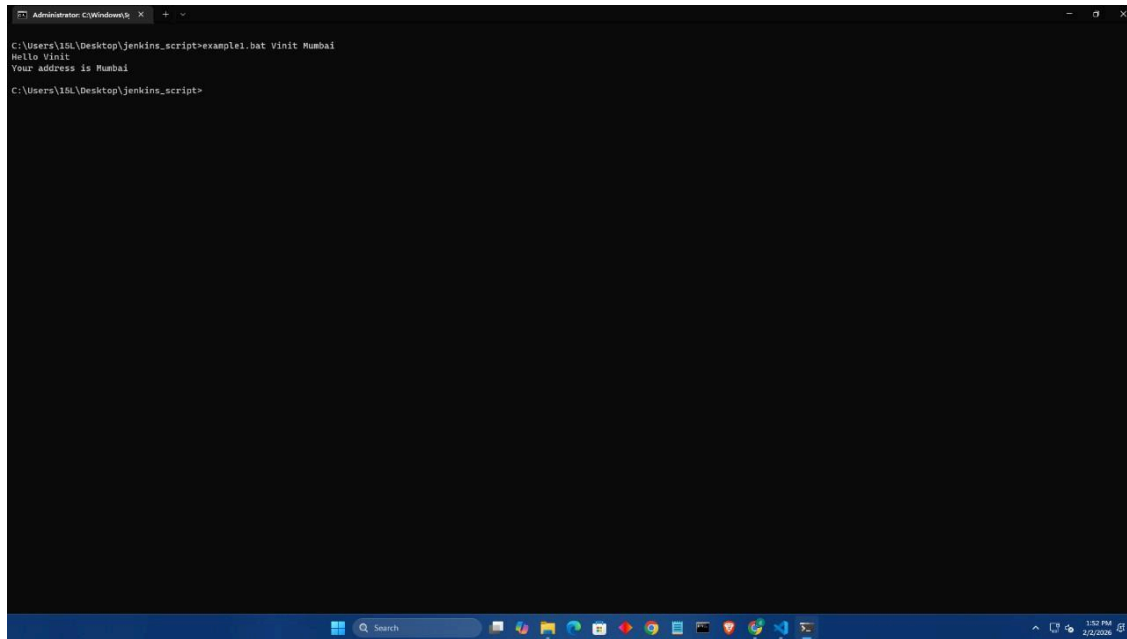
C:\ProgramData\Jenkins\jenkins\workspace\T13_63>echo Hello Jenkins
Hello Jenkins

C:\ProgramData\Jenkins\jenkins\workspace\T13_63>exit 0
Finished: SUCCESS
```

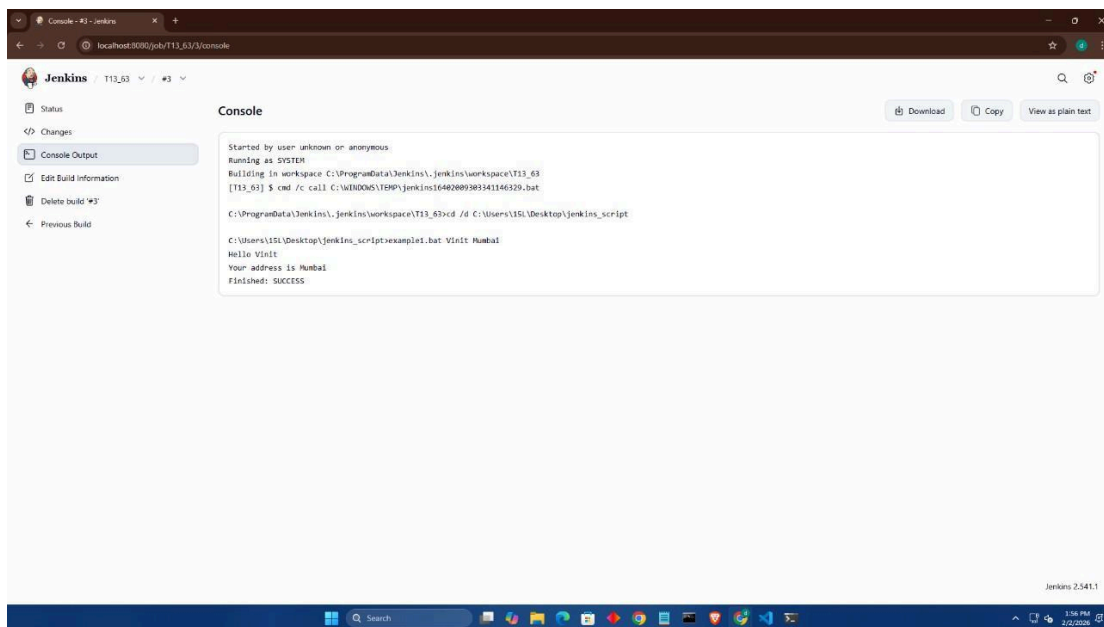
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Example 1



```
Administrator: C:\Windows\S...
C:\Users\ISL\Desktop\jenkins_script>example1.bat Vinit Mumbai
Hello Vinit
Your address is Mumbai
C:\Users\ISL\Desktop\jenkins_script>
```



Jenkins / T13_63 / #3

Status

Changes

Console Output

Edit Build Information

Delete build #3

Previous Build

Console

Download Copy View as plain text

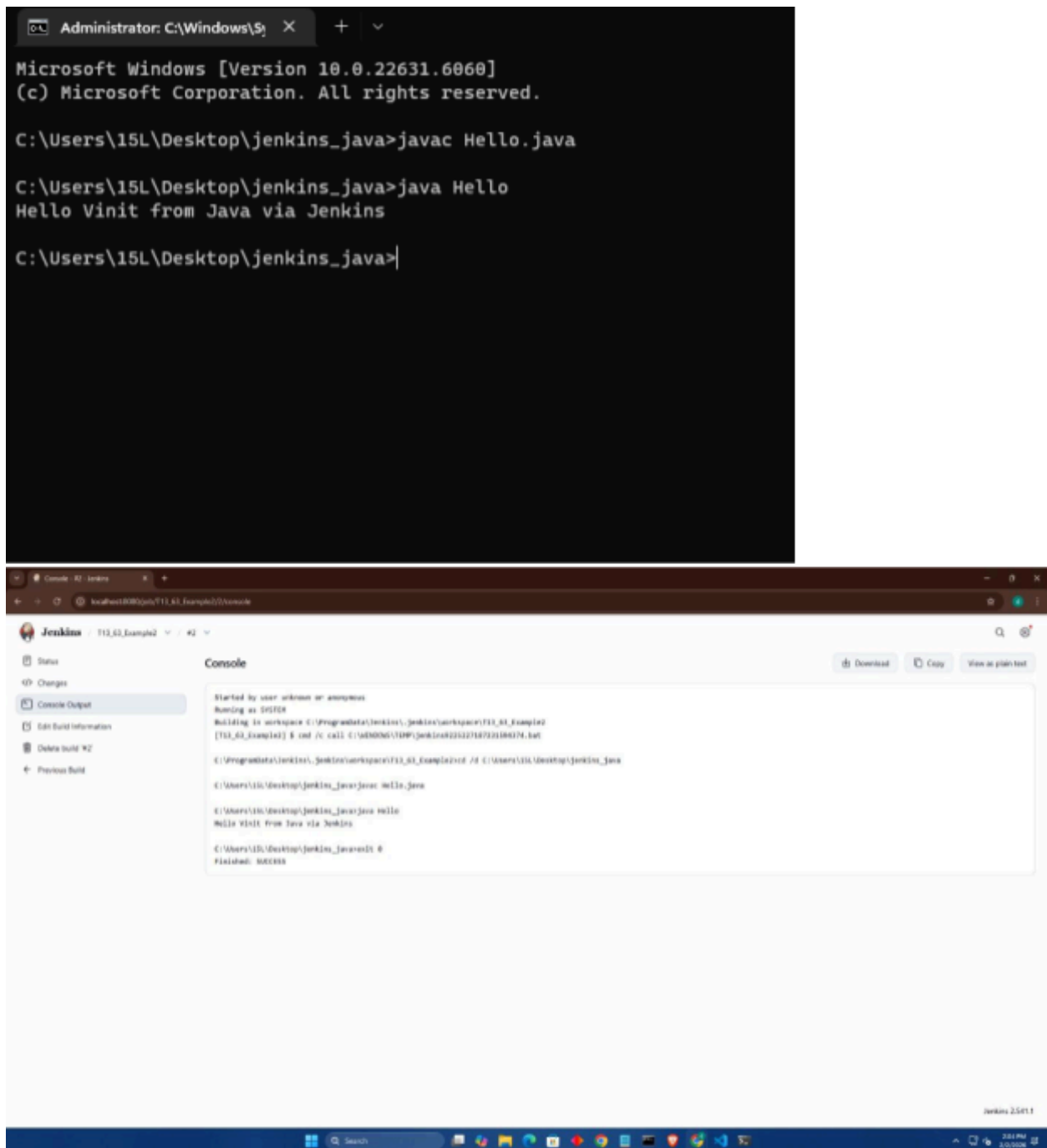
```
Started by user unknown or anonymous
Running as SYSTEM
Building in workspace C:\ProgramData\Jenkins\jenkins\workspace\T13_63
[T13_63] $ cmd /c call C:\Users\ISL\TEMP\jenkins164268938341166329.bat
C:\ProgramData\Jenkins\jenkins\workspace\T13_63>cd /d C:\Users\ISL\Desktop\jenkins_script
C:\Users\ISL\Desktop\jenkins_script>example1.bat Vinit Mumbai
Hello Vinit
Your address is Mumbai
Finished: SUCCESS
```

Jenkins 2.541.1

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Example 2



The image displays two screenshots. The top screenshot is a Windows command prompt window titled 'Administrator: C:\Windows\Sy...'. It shows the following commands and output:

```
Microsoft Windows [Version 10.0.22631.6060]
(c) Microsoft Corporation. All rights reserved.

C:\Users\15L\Desktop\jenkins_java>javac Hello.java

C:\Users\15L\Desktop\jenkins_java>java Hello
Hello Vinit from Java via Jenkins

C:\Users\15L\Desktop\jenkins_java>|
```

The bottom screenshot is a Jenkins web interface showing the console output for a build named 'T13_63_Example2'. The console output is as follows:

```
Started by user unknown or anonymous
Running as SYSTEM
Building in workspace C:\ProgramData\Jenkins\jenkins\workspace\T13_63_Example2
[T13_63_Example2] $ cd /c: call C:\WINDOWS\TEMP\jenkins8232757321085174.bat

C:\ProgramData\Jenkins\jenkins\workspace\T13_63_Example2>cd /d C:\Users\15L\Desktop\jenkins_java

C:\Users\15L\Desktop\jenkins_java>javac Hello.java

C:\Users\15L\Desktop\jenkins_java>java Hello
Hello Vinit from Java via Jenkins

C:\Users\15L\Desktop\jenkins_java>exit 0
Finished: SUCCESS
```

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Parameter

The screenshot shows the Jenkins web interface for configuring a project named 'Project t13_63_Example3'. The left sidebar contains navigation links: Status, Changes, Workspace, Build with Parameters, Configure, Delete Project, and Rename. The main area displays the configuration form with the following fields:

- ProfileName:** A text input field with the value 'Visit'.
- City:** A dropdown menu with the label 'Select your city'. The dropdown is open, showing a list of cities: Mumbai, Mumbai, Nagpur, Pune, Delhi, Bangalore, and Chennai. 'Mumbai' is selected.

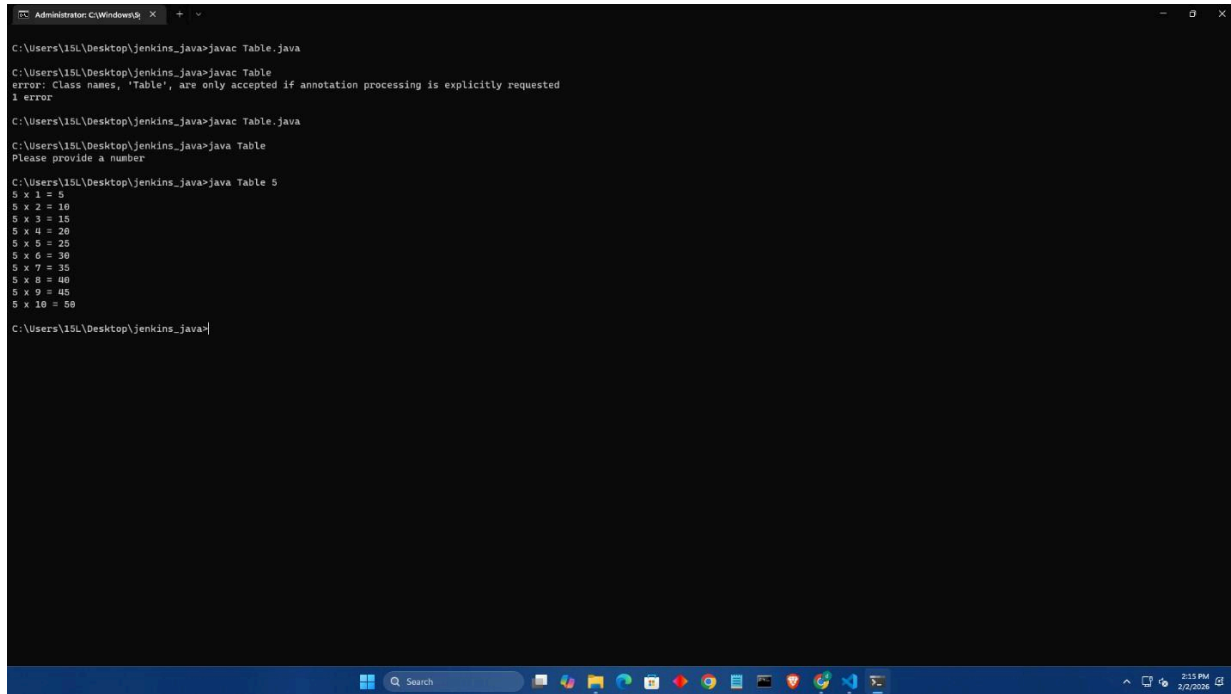
Below the configuration fields, there is a 'Builds' section showing 'No builds'.

This screenshot is identical to the one above, showing the Jenkins configuration page for 'Project t13_63_Example3'. In addition to the configuration fields, there are now two buttons at the bottom of the main area: a green 'Build' button and a grey 'Cancel' button.

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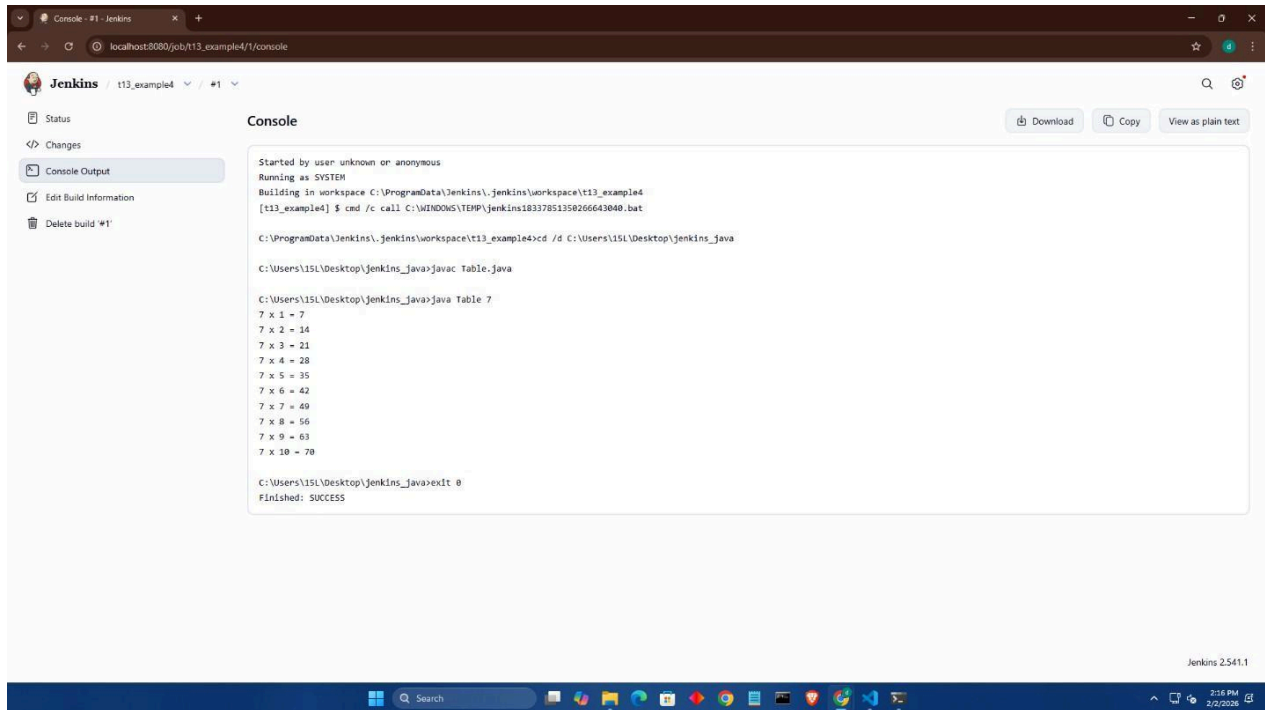
Table:



```
Administrator: C:\Windows\S...
C:\Users\151\Desktop\jenkins_java>javac Table.java
C:\Users\151\Desktop\jenkins_java>javac Table
error: Class names, 'Table', are only accepted if annotation processing is explicitly requested
1 error
C:\Users\151\Desktop\jenkins_java>javac Table.java
C:\Users\151\Desktop\jenkins_java>java Table
Please provide a number
C:\Users\151\Desktop\jenkins_java>java Table 5
5 x 1 = 5
5 x 2 = 10
5 x 3 = 15
5 x 4 = 20
5 x 5 = 25
5 x 6 = 30
5 x 7 = 35
5 x 8 = 40
5 x 9 = 45
5 x 10 = 50
C:\Users\151\Desktop\jenkins_java>
```

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The screenshot shows the Jenkins web interface in a browser. The console output for a build named 't13_example4' is displayed. The output shows the build starting as 'unknown or anonymous', running as 'SYSTEM', and building in the workspace 'C:\ProgramData\jenkins\jenkins\workspace\t13_example4'. The build script executed is 'call C:\WINDOWS\TEMP\jenkins1833785135826643040.bat'. The script changes the directory to 'C:\ProgramData\jenkins\jenkins\workspace\t13_example4>cd /d C:\Users\151\Desktop\jenkins_java' and then runs 'javac Table.java'. The output of the compilation is a multiplication table for 7, showing '7 x 1 = 7' through '7 x 10 = 70'. The build concludes with 'C:\Users\151\Desktop\jenkins_java>exit 0' and 'Finished: SUCCESS'. The Jenkins version 'Jenkins 2.541.1' is noted at the bottom right of the console area. The Windows taskbar at the bottom shows the time as 2:16 PM on 2/2/2026.

```
Started by user unknown or anonymous
Running as SYSTEM
Building in workspace C:\ProgramData\jenkins\jenkins\workspace\t13_example4
[t13_example4] $ cmd /c call C:\WINDOWS\TEMP\jenkins1833785135826643040.bat

C:\ProgramData\jenkins\jenkins\workspace\t13_example4>cd /d C:\Users\151\Desktop\jenkins_java

C:\Users\151\Desktop\jenkins_java>javac Table.java

C:\Users\151\Desktop\jenkins_java>java Table 7
7 x 1 = 7
7 x 2 = 14
7 x 3 = 21
7 x 4 = 28
7 x 5 = 35
7 x 6 = 42
7 x 7 = 49
7 x 8 = 56
7 x 9 = 63
7 x 10 = 70

C:\Users\151\Desktop\jenkins_java>exit 0
Finished: SUCCESS
```

Conclusion: Successfully Implemented Continuous Integration, install and configure Jenkins with Maven/Ant/Gradle to setup a build Job